EWP situations — review sheet (key)

You are on site and see each of the following situations. State whether or not it is allowed, and your specific reasoning.

	Situation	Acceptable yes/no	Document	Page	Detail/Chart	Correct Answer
1	Drawings call for 9.5" TJI S31. The flanges on site are 2.25" wide.	no	TJ-4510	2, 3		Correct width is 2.5". Not a TJI S31.
2	Drawings call for 11.875" TJI S31 @ 19.2" OC, 3/4" subfloor, 40+25psf, directly applied ceiling, continuous span = 18'-5". Client has decided to install a dropped ceiling rather than gypsum ceiling.	yes	TJ-4510	7		Maximum continuous span = 18'-5" with or without directly applied ceiling.
3	Drawings call for 16" TJI S31 roof joists @ 24"OC, 8/12 pitch. Contractor could only get 18".	no	TJ-4510	16, 17, 18		18" roof joist maximum slope = 3/12.
4	Drawings call for 3.5" x 3.5" Parallam 1.8E columns. Contractor substituted 3.5" x 3.5" Microllam 2.0E.	no	TJ-9500	2, 19		Microllam cannot be used for column applications.
5	Drawings call for 3.5" x 5.5" Timberstrand 1.3E lintel in plank orientation for interior sliding door. Carpenters drilled 1/4" holes in lintel to install the sliding door hardware.	no	TJ-9500	14		No holes in headers or beams in plank orientation.
6	Foundation is 36' wide. Drawings call for 11.875" TJI S31 @ 19.2" OC, 3/4" subfloor, 40+25psf, directly applied ceiling, continuous span = 18'-0". The longest joists available from the supplier are 20'.	no	TJ-4510	7		Maximum simple span = 17'-6" with directly applied ceiling.
7	Drawings call for 14" TJI S37. The flanges on site are 2.5" wide.	no	TJ-4510	2, 3		No such joist.
8	Drawings call for a 0.5/12 slope for a renovation using 14" roof joists. Contractor ordered 16" joists, so now the slope is 0.125/12.	no	TJ-4510	16, 17		Roof joist minimum slope = 0.25/12.
9	Drawings call for 8' high 3.5" x 3.5" Timberstrand 1.3E column on a wood plate. Contractor installed a base plate with the screws perpendicular to the wide face of the strands.	yes	TJ-9500	19		Column capacity on wood plate = 8,340# / capacity on base plate = 9,595#, OK. Screws installed correctly as per detail at bottom of page.
10	Drawings call for 3.5" x 11.875" Timberstrand 1.55E beam. The beam is 24' long with a column in the middle. Plumbers drilled a 3" diameter hole in the middle of the beam depth, directly above the centre of the interior column, then drilled a 2" hole 5" to the left.	no	TJ-9500	14		Maximum 3.625" hole is acceptable, 3" hole OK. The 2" hole must be 2 x diameter of the larger hole away, i.e. 2 x 3" = 6" away.
11	Drawings call for 11.875" TJI S31 @ 16" OC, 5/8" subfloor, 40+15psf, directly applied ceiling, simple span = 16'-11". Client has decided to install a dropped ceiling rather than gypsum ceiling.	no	TJ-4510	6		Maximum simple span without ceiling = 16'-10".
12	Drawings call for 16" TJI S31 roof joists @ 24" OC, 7/12 pitch, 25LL + 15DL, spanning 23'-9".	no	TJ-4510	16		7/12 = high slope. Maximum span = 23'-0".
13	Drawings call for 5.25" x 14" Microllam 2.0E instead of a girder truss to support the valley framing. The beam sits on a 2×6 wall. The factored reaction is 10,500#. The tapered end cut is 5.5" to stay under the sheathing.	no	TJ-9500	15		Maximum factored reaction for 5.5" heel = 9,480#. Beam requires a minimum heel height of 6.5" (10,870#).

	Situation	Acceptable yes/no	Document	Page	Detail/Chart	Correct Answer
14	Drawings call for 2 ply 1.75" x 11.875" Microllam 2.0E with a 12' clear span. Plumber drilled a 2" hole at 5', 7' and 9' in the middle of the beam depth.	no	TJ-9500	14		2" holes at mid-depth are OK. Holes spaced 2 × 2" = 4" edge to edge, OK. But the middle third of the beam = 4' to 8', not OK.
15	Drawings call for 9.5" TJI S31 @ 19.2" OC, 5/8" subfloor, 40+15psf, no directly applied ceiling, simple span = 14'-6". Client asks you if they will be happy with the floor performance.	65/35 chance	TJ-4510	6	Bold italic	Bold italic: 35% of population will find performance marginal to unacceptable.
16	Drawings call for 16" TJI S31 roof joists @ 24" OC, 5/12 pitch, 25LL + 15DL, spanning 23'-9". Carpenters bevelled the lower wall plate, installed shear blocking and used variable slope hangers and strap ties at the ridge beam.	yes	TJ-4510	16, 18, 19, 20	R1, H5S	Span OK (low slope 24'-0"). Bevelling lower wall plate OK. Shear blocking at lower wall OK. Sloped hangers and strap ties at ridge beam OK.
17	Drawings call for 5.25" x 14" Microllam 2.0E. The beam is side-loaded with a factored load of 400plf. The carpenter installed 3 rows of 0.128" x 3" nails @ 12" OC in the outside plies.	yes	TJ-9500	16	L17	Maximum allowable load = 650plf > 400plf, OK.
18	Drawings call for 14' long 11.875" TJI S31 @ 16" OC. Electrician drilled a series of 1" diameter holes in the middle of the web depth from 4' to 10' @ 6" OC.	yes	TJ-4510	13		1.5" holes are OK anywhere, therefore 1" holes OK. Spacing = 3" edge to edge, therefore 6" OC OK.
19	Foundation is 32' wide. Bearing wall is at 22'. Drawings call for 14" TJI S31 @ 16" OC, 5/8" subfloor, 40+15psf, directly applied ceiling, continuous span.	no	TJ-4510	6		10'/22' = 0.45 > 0.4, OK. Maximum span = 21'-2", not OK.
20	Drawings call for 14" S47 joists. Carpenters cut squash blocks 14 1/16" long.	yes	TJ-4510	10, 11	B2	Squash blocks to be 1/16" higher than joists.
21	Carpenter cut 6" seat for a bird's mouth for a 2×6 wall. Seat cut extends 1/2" past inside face of wall.	no	TJ-4510	20	BC	Bird's mouth cut cannot overhang inside face of plate.
22	Drawings call for 3.5" x 11.875" Timberstrand 1.55E beam. The beam is 24' long with a column in the middle. The end bearing requirement is 1.9". The cap on the interior column is 6" long.	yes	TJ-9500	7		Assuming the beam is continuous, bearing requirement at interior column = 2 × 1.9" x 1.25 = 4.8" < 6", OK.
23	Drawings call for 14' long 11.875" TJI S31 @ 16" OC. Joists are cantilevered 5" over brick, requiring an E6 detail. Electrician drilled 1/2" diameter holes in the cantilevers to run cable for exterior security cameras.	no	TJ-4510	13, 14	E6	Page 13: cannot cut holes in cantilever reinforcement.
24	Drawings call for 14" S47 joists. Carpenters cut web stiffeners 13 7/8" long.	no	TJ-4510	10	W	Maximum web stiffener length = 10 7/8".
25	Drawings call for 9.5" TJI S31. Joists are cantilevered 4" for brick and support the roof trusses. Backspan is only 24".	yes	TJ-4510	14		Minimum backspan = 2 x cantilever = 8", OK.